

NAGARJUN VINUKONDA

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SUMMARY

M.S. Robotics Engineering, Worcester Polytechnic Institute, MA — CGPA 4.0/4.0

B.E. Mechanical Engineering, VIT University, India — CGPA 3.84/4.0

- 5+ years of experience in learning-based robotic systems, **sim-to-real pipelines**, and **real robot deployments**.
- Specialization in behavior cloning, **flow matching**, **VLA** fine-tuning, and RL policy deployment.
- Fine-tuned **SmolVLA** on PushT, LIBERO, and Aloha sim tasks using LeRobot; full architecture understanding.
- Specialization in Perception, **ML/DL**, **Computer Vision**, **SLAM**, Motion Planning and Controls for Robot Autonomy.
- Built **large-scale ML evaluation pipelines** and VLA model hosting infrastructure at Toyota Research Institute.
- Model lifecycle and **post-training exposure**: SFT, PEFT, RLHF reward-model improvements, teacher-student distillation via FSDP, and CPT within production-scale LLM→VLM→VLA training systems.
- Built **distributed evaluation pipelines** with Data Distributed Parallel (DDP), Git CI automation, ONNX/Git SHA reproducibility, and long-tail data flywheel workflows.
- Led development of a 6-DOF surgical robot from POC to MVP at Neuro42 **within 8 months**.
- Worked on self-driving cars, mobile robots, manipulators, and surgical robots.
- Strong foundation in robot kinematics, dynamics, motion planning, controls, and sim environments.
- Proficient in Python, C/C++, PyTorch, ROS/ROS2, Docker, AWS, CUDA, Isaac Sim, MuJoCo, Gazebo, CARLA.
- **ML/DL experience**: 3D Object Detection Classification Transfer Learning Semantic Segmentation Transformers
- Published in [IJMET](#); Open-source contributor (PyQTgraph).

SKILLS

Robot Learning: Imitation learning, behavior cloning, flow matching, VLA/VLM fine-tuning, SmolVLA, LeRobot, Diffusion Policy, RL policy deployment

ML/DL Frameworks: PyTorch, TensorFlow, ONNX, TensorRT, Hugging Face, W&B, JAX (familiar)

Simulation: Isaac Sim, MuJoCo, Gazebo, CARLA, CarMaker, CALVIN, LIBERO, RoboCasa, Rviz

Robotics & Control: ROS/ROS2, FK/IK, trajectory generation, motion planning, EKF, SLAM, sensor fusion, PCL, OpenCV

Programming: Python, C/C++ (C++14), MATLAB, Bash, Linux, Docker, AWS, Git CI/CD, CUDA

Hardware: Jetson Xavier/Orin, NI-DAQ, PLC, encoders, pneumatic actuators, Arduino, SolidWorks

WORK EXPERIENCE

Senior Software Engineer — Machine Learning, Toyota Research Institute (Contract) Sep 2025 – Present

Tools: Python, PyTorch, ROS2, LeRobot, Isaac Sim, MuJoCo, ONNX, Wandb, Hugging Face, Docker, AWS, Git CI

- **VLA Production Pipeline Exposure**: Worked with VLA Foundry team got exposure to — a production-scale LLM→VLM→VLA training and evaluation system running on SageMaker across multi-node GPU clusters, covering SmolVLA, Diffusion Policy, and ManiFlow across sim and real hardware.
- **SmolVLA Hosting & sim Harness**: Developed model hosting wrappers and evaluation harness for SmolVLA policy inference via LeRobot across Isaac Sim, MuJoCo, CALVIN, LIBERO, and RoboCasa — supporting policy rollout benchmarking and sim-to-real gap analysis across multiple manipulation tasks.
- **Policy Diagnostics & Logging**: Designed and merged sparse structured logging infrastructure for VLA_foundry inference policies with Rerun and W&B integration — enabling runtime debugging and systematic diagnostics of policy rollouts on real and simulated robots.
- **Large-Scale Data & Eval Pipelines**: Built ML evaluation pipelines for AV and robotics VLA models; ONNX import/export with Git SHA tracking; Git CI-integrated automated evaluation pipeline using Data Distributed Parallel (DDP) for End-to-End model inference and tracking on the Waymo dataset.
- **Multi-Camera Perception**: Developed panoramic image stitcher via feature matching, homography, and extrinsic projection for multi-camera robot sensor setups.
- **Post-Training & Model Optimization Exposure**: Within that pipeline, gained exposure to post-training via SFT and PEFT for optimization and RLHF for scene-understanding reward-model improvements, alongside sim-to-real gap analysis via LBM evaluation.
- **Distillation Infrastructure & CPT**: Exposure to teacher-student distillation and infrastructure pipeline development via FSDP; continued pre-training (CPT) for domain knowledge.

- **Data Flywheel & Long-Tail Events:** Exposure to the complete End-to-End pipeline — from collecting long-tail event data flywheel through Training, Evaluation, and Inference.
- **BEV Perception & Intent Prediction:** Working on multi-model outputs for the vehicle pull-over system and inactive-vehicle detection using BEV representations, with intent regression using transformer models.

Robotics & ML Engineer, [Neuro42](#) | Medical Robotics

Oct 2023 – Sep 2025

Tools: C++, ROS2, Python, OpenGL/GLSL, PyQT5, IsaacSim, PyTorch, SolidWorks, NI-DAQ, PLC, Linux

- **Team Lead — Surgical Robot POC to MVP:** Led cross-functional team (SW, EE, ME) to take a 6-DOF pneumatic surgical robot from 2-DOF POC to MVP within 8 months.
- **Real-Time Control on Physical Hardware:** Developed C++ control loops for multi-axis pneumatic motor actuation via PLC and NI-DAQ; tuned closed-loop control with encoder feedback — all validated on the physical surgical robot.
- **Kinematics & Trajectory Generation:** Derived FK/IK for 3—6 DOF arm; implemented goal-directed waypoint trajectory generation with obstacle-aware motion execution for full end-to-end autonomous operation.
- **HIL Digital Twin:** Built bidirectional HIL pipeline via ROS2 — sim commands drive real actuators while encoder feedback closes the loop, enabling closed-loop hardware-software validation.
- **3D Simulator & Software Stack:** Built the 3D simulation and visualization from scratch using OpenGL/GLSL (mesh/surface rendering of reconstructed volumes); architected the entire [Robot Software Stack](#) from Qt UI to backend PLC communications.
- **Perception / ML:** 3D reconstruction and anomaly detection for brain MRI images using 3D UNet; co-registration on high-field MRI data.

Perception Engineer, [Forterra](#)

Feb 2022 – Oct 2023

Tools: C++, ROS, Python, OpenCV, CUDA, TensorRT, Ceres Solver, Jetson Xavier/Orin, AWS, Multi-threading

- **Sensor Calibration Pipelines:** Developed intrinsic (pinhole/fisheye) and extrinsic calibration pipelines (LiDAR-to-Camera, LiDAR-to-RADAR) using ICP and RANSAC; built GUI for single-shot multi-camera calibration; deployed results to AWS.
- **Sensor Fusion & Localization:** Built EKF-based multi-sensor fusion module for LiDAR and Camera; improved AMCL localization; developed multi-obstacle track management system.
- **Quantization & TensorRT Optimization:** Applied Post-Training Quantization (PTQ) and Quantization-Aware Training (QAT) for mixed-precision inference optimization; deployed a DenseNet-UNet **Semantic Segmentation** model trained via transfer learning using TensorRT on Jetson hardware.
- **Embedded & Networking Systems:** Configured Jetson Xavier device trees on kernel BSP for GMSL cameras; built GStreamer multi-camera pipeline with UDP sink for remote visualization; performed latency and network throughput analysis.
- **Computer Vision:** Trailer angle detection via Homography; camera ISP quality testing (auto exposure, noise reduction, edge enhancement); LiDAR projection onto camera, **obstacle detection & semantic segmentation**
- **Testing & Validation:** Developed test setups, accuracy tests, and noise filtering for LiDAR, RADAR, and Camera calibration pipelines; created Bash scripts for GUI build, data collection automation, and PCAP file generation.

ADAS Simulation Engineer, [Forterra](#)

Feb 2022 – Jul 2022

Tools: C++, ROS, CarMaker, TruckMaker, OpenDRIVE, OUSTER LiDAR, SIL/HIL

- **Simulation Environments:** Developed high-fidelity AV simulation environments from satellite imagery and topography data; handled UTM↔Lat/Long coordinate transforms, converted scene formats to OpenDRIVE/OpenStreetMap, and integrated LiDAR drivers and ground-truth sensor models for SIL/HIL testing.

Motion Planning Engineer, [Thordrive](#)

Sep 2021 – Feb 2022

Tools: C++, ROS, Eigen, Dijkstra, KD-Tree, Rviz

- **Motion Planning & Controls:** Built full AV route planner from scratch to Lvl3 autonomy; tuned Hybrid-A* for constrained trajectory scenarios; tuned Pure Pursuit and Stanley controllers on vehicle hardware.

Robotics Navigation Intern, [Midea Group Emerging Technologies](#)

Jul 2021 – Sep 2021

Tools: Python, Raspberry Pi, TensorFlow, PyCoral TPU, Linux, Pytest

- Trained object detection NNs on COCO/KITTI; deployed inference on Google Edge PyCoral TPU; computed mAP, IOU, precision, recall.

Research Assistant — Social-Aware Navigation, [HiRo Lab, WPI](#)

Sep 2020 – May 2021

Tools: C++, ROS, Gazebo, Kalman Filter, AMCL, GMapping, Eigen, OpenCV, CI/CD

- **SLAM & Navigation:** Implemented GMapping + AMCL-based SLAM; designed ORCA velocity obstacles for dynamic human-obstacle avoidance; built Kalman filter-based obstacle tracker with social constraints.

PROJECTS

VLA Policy Training & Evaluation | LeRobot, PyTorch, W&B, Isaac Sim, MuJoCo

- Fine-tuned SmoIVLA across PushT and LIBERO Spatial's 10-task manipulation suite; tracked flow-matching loss, gradient norms, and policy rollout performance using Weights & Biases.
- Developed simulation environments and evaluation harnesses across CALVIN, LIBERO, and RoboCasa, integrating policy inference, structured logging, and rollout analysis.
- Enabled local training and evaluation within 8GB VRAM through batch-size tuning and memory configuration; implemented asynchronous inference and action-chunk aggregation to reduce execution jitter.

SmoIVLA Pick-and-Place & Sim-to-Real Debugging | PyTorch, LeRobot, Sim-to-Real Gap

- Evaluated pick-and-place policies for cubes, cylinders, tetrahedra, and octahedra; diagnosed orientation-sensitive alignment and insertion failures on harder shapes during sim-to-real transfer.
- Fine-tuned and debugged with targeted demonstrations and recovery data, adjusted observations for camera mismatch, and shortened action execution horizons for more frequent replanning near insertion.

TurtleBot Navigation using DQN — ROS, Python, PyTorch, CUDA

- Implemented DQN and Dueling DQN in PyTorch with experience replay and target networks; trained and evaluated across Gazebo stages with static and Dynamic moving obstacles — RL policy trained and deployed on physical robot hardware.
- Designed the navigation MDP: a 26-D observation of downsampled LiDAR ranges, goal heading, and goal distance (28-D with nearest-obstacle range and bearing), and 5 discrete steering actions at fixed linear velocity.
- Shaped the reward around goal-heading alignment and an exponential distance-to-goal term, with proximity penalties for nearby obstacles and terminal goal/collision rewards tuned per stage.

Isaac Sim Franka Pick-and-Place: Teleop → Mimic → SmoIVLA Pipeline

- Teleoperated a 7-DoF Franka Panda in Isaac Sim via keyboard SE(3) control to collect seed demonstrations of cube-to-bin manipulation, recorded to HDF5 through Isaac Lab's record_demos pipeline; expanded them into a larger trajectory set using Isaac Lab Mimic's subtask-aware replay with pose randomization.
- Converted HDF5 → LeRobot dataset format, fine-tuned SmoIVLA on a cloud L40S, and closed the loop by evaluating the checkpoint with inference back in the same Isaac Sim scene.

Robot Arms Teleoperation with Shared Autonomy — ROS, Python, OpenCV, Baxter

- Implemented shared autonomy for 7-DOF Baxter arms with fully/semi-autonomous modes and joystick controls; vision-based grasp manipulation with RealSense camera calibration — teleoperation and demonstration collection for robot learning.

Behaviour Cloning for Autonomous Vehicle — CARLA, TensorFlow, Keras, GPU

- Generated real-time demonstration data and trained CNN to clone EGO driving behaviours in CARLA — end-to-end imitation learning pipeline from data collection to policy deployment.

Dynamic Step Planning with LiDAR for Human Exoskeleton — ROS, PCL, C++, SickTim LiDAR

- RANSAC-based staircase detection from LiDAR point clouds;
- Joint trajectory generation for Exo-Legs using **Dynamic Motion Primitives(DMPs)** with human gait data;
- FK/IK for 3-DOF exoskeleton leg for stair climbing and generated step trajectories

Pizza Slice Orientation Estimation — PyTorch, Self-Supervised Pretraining, Docker

- Built a two-stage angle-regression pipeline: self-supervised convolutional autoencoder pretraining on unlabelled images, then fine-tuned the transferred encoder with regression head.

Indoor 3D Mapping using RGBD Camera — ROS, C++, Gazebo, RTAB-Map

- Full RGB-D SLAM pipeline using RTAB-Map with loop closure and pose graph optimization; 2D occupancy grids and dense 3D point cloud maps in Gazebo.

Behaviour Planning for Autonomous Driving — MATLAB, Simulink, CARLA, C++

- Vehicle behaviour prediction with sensor fusion; nonlinear MPC tuned for bicycle kinematic model trajectory tracking.

3D Object Detection — Modified Frustum PointNets — Python, TensorFlow, GCP

- 3D detection on KITTI dataset; LiDAR-Camera fusion for pedestrian spatial localization.

Route Planning on Open Street Map — C++14, Python, OSM

- Comparative C++ implementation of BFS, DFS, A*, Dijkstra, and RRT* planning algorithms; A* on OSM data with IO2D rendering.

TurtleBot Path Tracking using PID Controller — ROS, C++, Turtlebot 2.0

- Hardware and software PID control for single and multi-goal waypoint tracking on a physical robot.

Video Stream TCP Comm using WebRTC — Python, OpenCV, aiortc, Multiprocessing, Docker, Pytest

- Multi-processing thread for image coordinate computation; Docker image implementation; unit tested server and client pipelines.

Traffic Signal Detection System — OpenCV, Python, SVM, RealSense D435

- Real-time YOLOv3 detection at 95% accuracy; SVM for signal status recognition; RealSense calibration with SIFT/SURF feature extraction.

Sensor Fusion and Tracking of IMU & GPS using GTSAM — Python, GTSAM, IMU, GPS

- Fused real-time IMU and GPS data using factor graph optimization (GTSAM) to produce a unified state estimate from accelerometer, gyroscope, and location data.

Transfer Learning with VGGNet — Keras, Python, TensorFlow, GPU

- Experimented with VGGNet on ImageNet dataset; comparative analysis against GoogleNet and ResNet.

Lane Detection using Camera Sensor — Python, Canny Edge, Hough Transforms, OpenCV

- Real-time lane detection and tracking using Canny edge detection and Hough transforms on highway lanes.

Human-aware Navigation with RVO and Artificial Potentials — ROS, Python, RVO, Gazebo

- Combined Artificial Potential Fields with RVO for obstacle avoidance; set-point controller for lateral waypoint tracking in Gazebo.

Model Predictive Controller for Autonomous Vehicles — C++, CARLA

- Designed and tuned MPC for lateral and longitudinal control of an autonomous car across given tracks and goal locations in CARLA for rear-end bicycle model.